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CNOT Circuit Identities

Video: Building a CNOT Gate Identity Circuit with Hadamard Gates
3 min

Video: Creating Circuit Equivalence: Reversing CNOT Gates in Qiskit
3 min

Graded Assignment: Assessment 11
Submitted

Your grade: 100%

Your latest: 100% • Your highest: 100% • To pass you need at least 80%. We keep your highest score.

Assessment 11

Next item →

1. Which of the following are Hadamard gates?

1 / 1 point

☒ It can create an equal superposition state from

coach

|0⟩

Ready to review what you've learned before starting the assignment? I'm here to help.

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☒ Correct

Correct! The Hadamard gate transforms the

Assignment details

|0⟩

state into an equal superposition of

Due

Dec 23, 11:59 PM IST

Attempts

Unlimited

Retry

Submitted

Dec 31, 2:10 AM IST

and

|1⟩

Your grade

To pass you need at least 80%. We keep your highest score.

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☐ It leaves the

100%

|1⟩

state unchanged.

☐ It's a two-qubit gate.

Like

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☒ It can be represented by the matrix

$$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

☒ Correct

Correct! This is the matrix representation of the Hadamard gate.

2. What is the primary function of the X gate in quantum computing?

1 / 1 point

☒ To flip the state of a qubit from |0⟩ to |1⟩ or vice versa

☐ To swap the states of two qubits

☐ To measure the state of a qubit

☐ To entangle two qubits

☒ Correct

Correct! The X gate flips a qubit's state, acting as a quantum equivalent of the classical NOT gate.

3. Identify the correct statements about the impact of the X gate on a qubit's state.

1 / 1 point

☒ It acts as a quantum equivalent of the classical NOT gate

☒ Correct

Correct! The X gate is considered the quantum analog of the classical NOT gate.

☒ It flips the state of a qubit from |0⟩ to |1⟩

☒ Correct

Correct! The X gate flips a qubit's state from |0⟩ to |1⟩.

☐ It measures the state of a qubit

☐ It creates a superposition state

☒ It flips the state of a qubit from |1⟩ to |0⟩

☒ Correct

Correct! The X gate also flips a qubit's state from |1⟩ to |0⟩.

4. What are the characteristics of circuits that are considered equivalent?

1 / 1 point

☒ They can be transformed into each other without changing the output.

☒ Correct

Right! Equivalent circuits can be transformed into each other without altering the output.

☐ They use the same types of gates.

☒ They produce the same output for all input states.

☒ Correct

Correct! Equivalent circuits must produce the same output for all input states.

☐ They must contain at least one entangling gate.

☐ They have the same number of gates.

5. Which gate is used to transform a qubit into an equal superposition of |0⟩ and |1⟩?

1 / 1 point

☒ Hadamard gate

☐ CNOT gate

☐ Toffoli gate

☐ Pauli-X gate

☒ Correct

Correct! The Hadamard gate is used to create an equal superposition of |0⟩ and |1⟩.